

BEN CARNES

Software Developer — Statistics Major, University of Waterloo

[GitHub](#) — [LinkedIn](#)

[Portfolio](#)

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, HTML/CSS, SAS, PHP

Data: PyTorch, LangChain, Azure OpenAI, Qdrant

Tools: React, PySide6, Git, Bash, Linux, Docker, Jupyter

Cloud: Azure, CI/CD, OWASP ZAP

EXPERIENCE

Software Developer Intern

Jan 2026 – April 2026

Statistics Canada

Ottawa, ON

- Implemented 4 reusable modules for generating derived variables for an internal data transformation platform.
- Developed and debugged custom Blockly blocks for visual data transformation workflows.
- Created a working automated OWASP ZAP security scanning reference DevOps pipeline, providing teams with a reusable example for CI/CD vulnerability checks.
- Refactored 3 legacy SAS workflows to Python, improving maintainability and long term extensibility.
- Fixed bugs across internal tools and helped teammates debug implementation issues.

Full Stack Software Developer Intern

Apr 2025 – Aug 2025

LC Institute

Waterloo, ON

- Designed an end-to-end RAG chatbot using Azure OpenAI and Qdrant, giving the nonprofit a natural language Q&A system from scratch.
- Built the chatbot backend orchestrator for handling user queries, then deployed it with Docker and Azure Container Apps and added content safety checks.
- Automated monthly vector ingestion with Azure Container Instances and documented the system architecture.
- Kept projected cloud costs under \$200/month while planning for up to 500 concurrent users.
- Deployed and secured the chatbot's vector database to support reliable backend search.
- Integrated Microsoft Entra ID authentication and secure query routing into the backend orchestrator for the user-facing WordPress UI, connecting users to the chatbot.

PROJECTS

Gamegrab | PySide6, EasyOCR, PyTorch, C++/DLL, DeepL API

- Built a desktop OCR + translation tool for PC games, with a capture-to-translation workflow for Japanese dialogue, menus, and on-screen UI.
- Added cross-resolution capture support and a simple 4-button interface for capturing, refreshing, reading, and translating text.

Autonomous Driving Simulation | C++, ROS2

- Implemented 4 ROS2 nodes for a robot sim that used local costmaps, global map memory, and A* path planning to navigate around obstacles.
- Wired up the planner, map memory, and control nodes through ROS2 topics and frame IDs so the robot could replan and move toward a goal.

WGCapture | C++, DLL, Direct3D, Python, PyPI

- Published a lightweight Python screen-capture library on PyPI, backed by a custom C++/DLL implementation.
- Used Direct3D frame-pool capture to support fullscreen, borderless, per-window, and high-DPI capture scenarios.

CLUBS & ACTIVITIES

WATonomous | Autonomous Vehicles Student Design Team

May 2026 – Present

EDUCATION

University of Waterloo

Bachelor of Mathematics

2028

Waterloo, ON